

Respiration and Photosynthesis Exam Questions

studyclix

Name:

Oisín Argue

Question 1

- (b) (i) Name the part of the cell where photosynthesis takes place.

~~Mitochondria~~ Chloroplast ✓

- (ii) Name the part of the cell where respiration takes place.

Mitochondria ✓

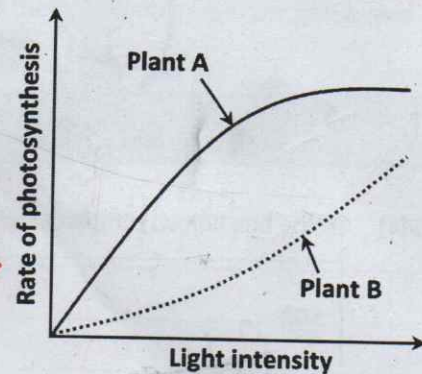
- (iii) Glucose reacts with gas Z in the mitochondrion. Identify gas Z.

~~Carbon dioxide~~ Oxygen ✓

- (c) The graph on the right shows the relationship between light intensity and the rate of photosynthesis for two different plants, A and B.

- (i) Explain the relationship between light intensity and the rate of photosynthesis for plant B.

As light intensity increases, the rate of photosynthesis also increases. ✓



- (ii) Which plant, A or B, is better adapted to a habitat where there is low light intensity? Justify your answer.

~~B is better adapted because at low light intensity the rate of photosynthesis is higher.~~

A is better adapted because if there is too much light, the rate of photosynthesis decreases. ✓

Plant A because at low light levels, there is a higher rate of photosynthesis.

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Question 2

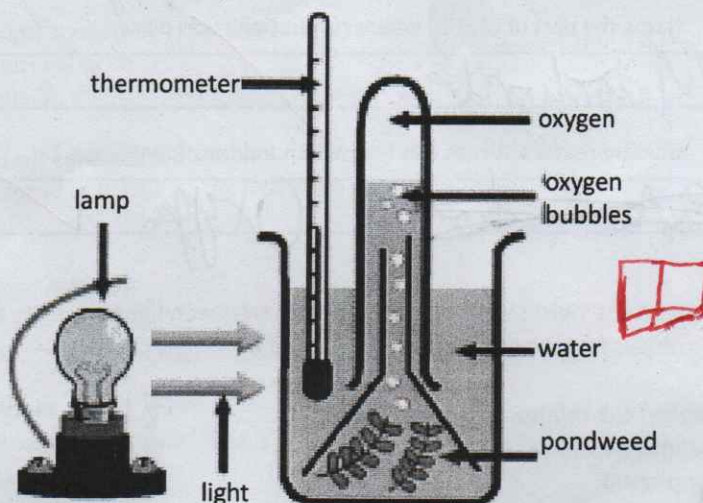
Question 14

(45 marks)

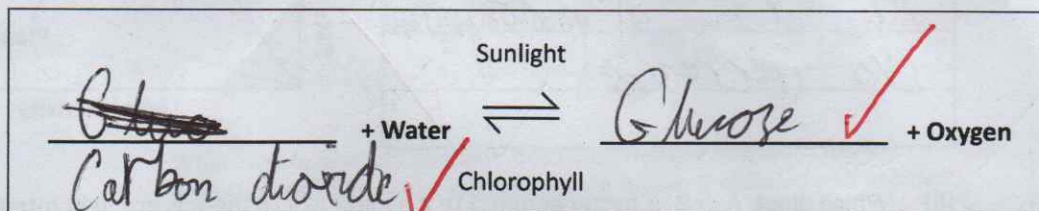
Under the supervision of their teacher, a group of students were investigating the effect of light intensity on the rate of *photosynthesis*.

They set up their apparatus similar to that shown in the diagram below.

The students measured the distance between the light source and the pondweed.



- (a) In the box below, complete the word equation that summarises photosynthesis.



- (b) Outline a suitable hypothesis that the students could have formed for this investigation.

If the light intensity increases, then the rate of photosynthesis will also increase ✓

- (c) Name one piece of apparatus that the students could have used to measure the distance between the light source and pondweed.

Metre stick ✓

- (d) Name the variable that was measured in this investigation.

Rate of photosynthesis ✓

- (e) Name the variable that was changed in this investigation.

Light intensity ✓

- (f) The students performed this investigation three times. Give a brief explanation for this.

So that their results could be trusted in case one investigation had something wrong with it ✓
Accuracy

- (g) The results in the table below were obtained during the investigation. Comment on the relationship between the distance of the light source from the pondweed and the average number of oxygen bubbles per minute.

Distance (cm) from light source	Average number of oxygen bubbles in 1 minute
10	12
20	10
30	7
40	6
50	4
60	2

Comment:

The closer the light is to the pondweed, the more oxygen bubbles there will be. ✓

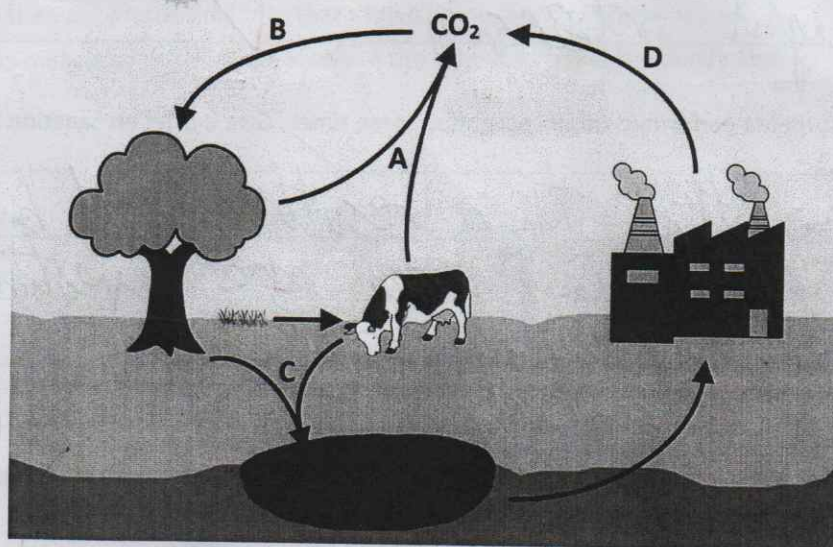
As the distance of the light source decreased, the number of oxygen bubbles increased

Question 3

Question 11

(30 marks)

The diagram below shows how carbon moves into and out of the air and soil. This is part of the carbon cycle.



(b) During process B, plants take in carbon dioxide from the air to make food.

(i) Name process B.

Photosynthesis ✓

(ii) Complete the chemical equation below for process B.

You may use either the name or the chemical formula for each substance to complete the equation.

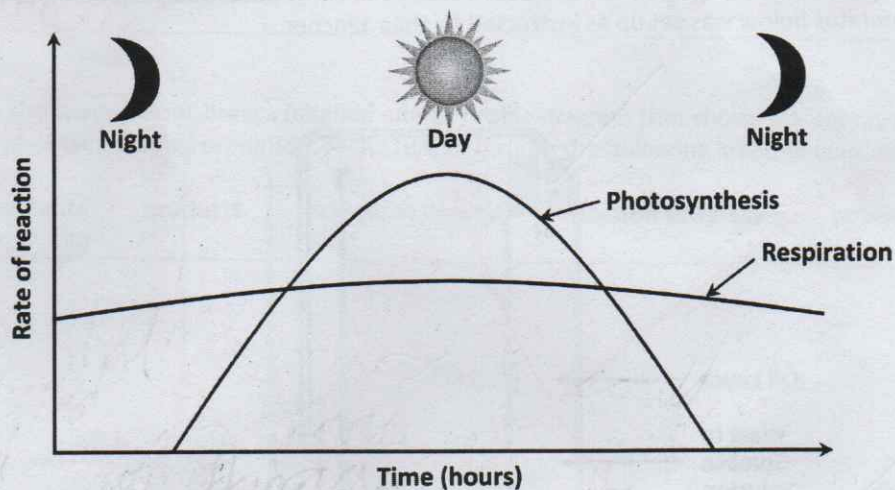
Carbon dioxide (CO_2) + water ✓ → Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) + Oxygen ✓

(iii) Name the green chemical found in plants which allows process B to happen.

Chloroplast X Chlorophyll

Question 4

- (f) The graph below shows how the rates of photosynthesis and respiration for a plant varied over a period of 24 hours.



- (iii) Explain why the rate of respiration remains relatively constant over 24 hours, compared to the rate of photosynthesis.

Because, the respiration requires the product of photosynthesis, which can be stored, but photosynthesis requires constant new sunlight ✓

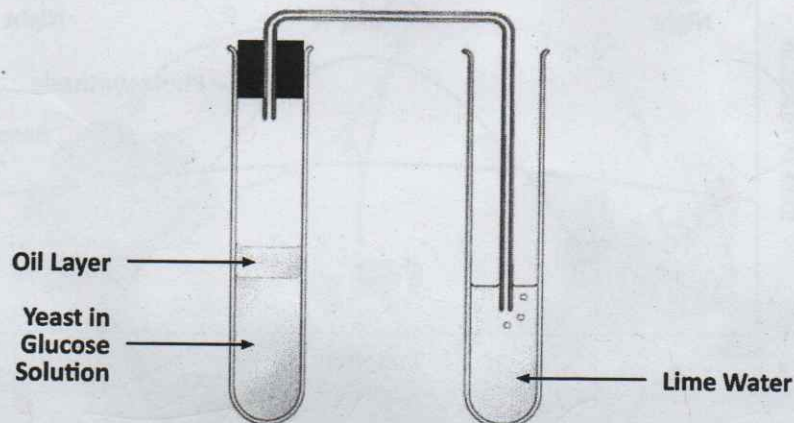
Photosynthesis depends on light, while respiration occurs at a constant rate

Question 5

Question 11

(30 marks)

Two students were studying the conditions necessary to show anaerobic respiration of yeast cells. The apparatus below was set up as instructed by their teacher.



- (a) Outline two safety precautions necessary to carry out an investigation such as this.

To keep away from electrical outputs ✓
To tie back hair so as not to interfere. ✓

- (b) Complete the word equation that summarises the process of aerobic respiration by adding in the names of the missing chemical compounds.

Oxygen + Glucose $\Rightarrow$ Water + Carbon Dioxide

- (c) Prior to the investigation, the glucose solution was heated to approximately 37°C to activate the dried yeast. Name one piece of apparatus that could have been used to heat this solution and one piece of apparatus that could be used to confirm the temperature was at 37°C.

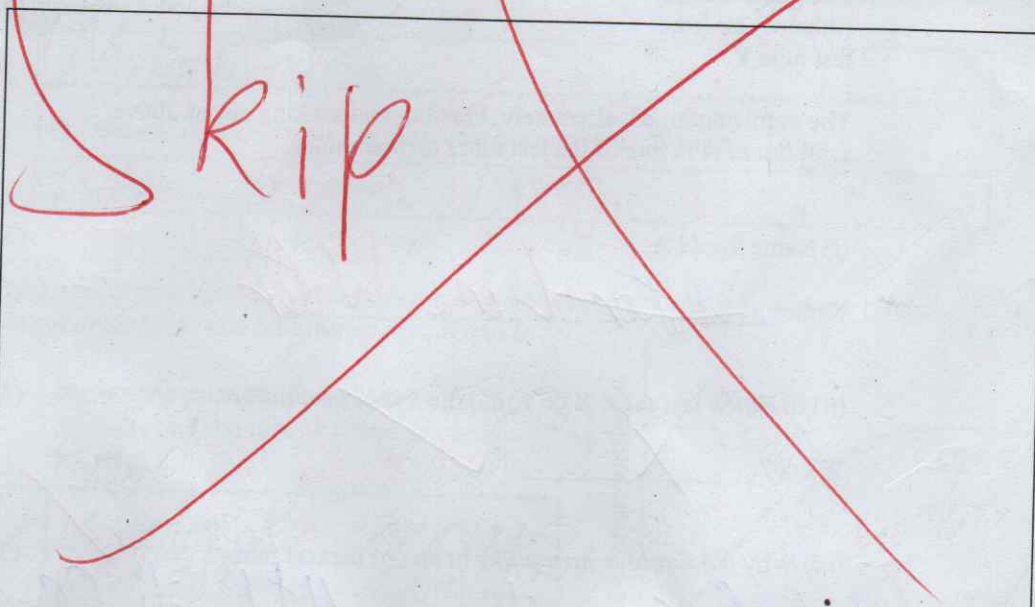
Heat source (apparatus)	Hot plate ✓
Temperature monitoring (apparatus)	Thermometer ✓

- (d) Respiration is an example of an exothermic reaction. Explain the underlined term.

Exothermic reactions give off heat

- (e) In the space below, draw a **labelled** energy profile diagram that shows the energy change that occurs during respiration. Make sure to include the following labels in your diagram:

reactants products activation energy reaction progress potential energy



Question 6

- (f) An earthworm takes in and uses up a certain gas during respiration.

- (i) Name the gas the earthworm uses up during respiration.

Oxygen

- (ii) Name a gas the earthworm produces during respiration.

Carbon dioxide



Question 7

- (e) Supplies of food (glucose) and oxygen are needed for aerobic respiration. During this process energy and two waste products are released.

- (i) Name one of these waste products.

Water Carbon dioxide

- (ii) Name the chemical used to test for the waste product you have named.

~~Carbon dioxide~~ Lime water

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examiner
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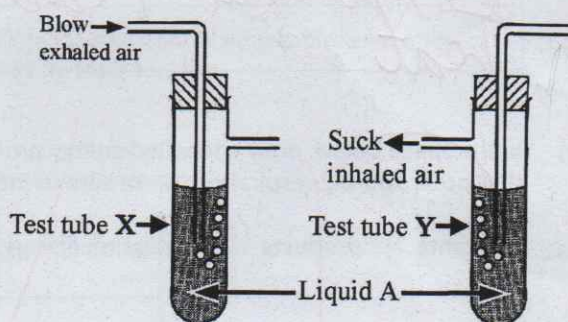
(1) (2)

Question 8

- (b) The diagram shows the apparatus used by a pupil when performing an experiment in a school laboratory.

The pupil blew (exhaled) air into test tube X.

The pupil sucked (inhaled) air from test tube Y.



The pupil continued, alternately, blowing and sucking air, as above, until **liquid A** in **one** of the test tubes **turned milky**.

- (i) Name **liquid A**.

Name

~~Water~~ **Limewater**

(3)

- (ii) In **which** test tube, X or Y, did the **liquid turn milky**?

Which?

X

(3)

- (iii) Why did **liquid A turn milky** in **one** of the test tubes?

Why?

Because it was polluted with CO₂

(3)

- (iv) What **conclusion** can be made from the **result of this experiment** regarding the **difference in composition between exhaled and inhaled air**?

Conclusion?

There is CO₂ in exhaled air and none in inhaled

(3)

- (v) Complete the **word equation**, below, for **aerobic respiration**.

Food +

oxygen

→

carbon dioxide

+

energy + water

(6)